

Capability-Matched Likelihood for Forensic Audits of Reasoning-LLM Distillation

Jie Ni, Chenning Zhang, and Xinting Zhang

Abstract—Reasoning large language models can be copied by distilling chain-of-thought traces into cheaper students, but provenance tests must separate descent from ordinary task competence. We study a forensic audit in which the suspect is observed only through output traces, candidate teachers are available for authorized likelihood scoring, and the suspect base is known or declared. The common base-relative likelihood surplus is confounded by candidate capability: in our GSM8K audit it false-alarms on a human-reference student (empirical false-positive rate, FPR, 0.81) and misses same-family distillation (area under the receiver-operating characteristic curve, AUROC, 0.13). Capability-matched likelihood (CML) instead compares suspect traces with a same-base non-distilled reference under the same candidate teacher. On three-seed Qwen2.5-7B-Instruct GSM8K students distilled from three teachers, CML reaches trace-level AUROC 0.997, true positive rate at 1% FPR 0.991, and empirical FPR 0.2%; it also recovers same-family distillation (AUROC 0.998). A single-seed MATH stress test shows the same pattern (AUROC 0.993). Multi-Task Calibration Reference (MTCR) improves sibling-teacher attribution from 0.672/0.658 to 0.894/0.875. Aggregate stress summaries further test adaptive trace transformations, cross-base and cross-task transfer, open-set decoys, and student-level confidence intervals: CML+MTCR retains AUROC at least 0.930 across the reported adaptive sweeps, the cross-base/task matrix has minimum AUROC 0.912, and open-set false attribution stays at or below 2.0%.

Index Terms—Model provenance, knowledge distillation, intellectual-property forensics, large language models, hypothesis testing, reasoning models.

I. INTRODUCTION

REASONING large language models trained to emit explicit chain-of-thought (CoT) before answering have reset the cost frontier of high-end machine intelligence. Their CoT traces are a high-value supervision signal: an adversary with only query access can harvest the teacher’s trajectories and supervised-fine-tune a far cheaper student that inherits much of the teacher’s reasoning skill. This distillation-from-traces recipe, popularized by the open release of R1-style distilled checkpoints, turns a teacher’s most valuable asset into a liability—the student can be rebranded, sold, or served behind an API as an ostensibly independent model, and the owner has no calibrated instrument for supporting a copy claim.

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A. Prevention is preempted; provenance is the live problem

Stopping distillation by modifying teacher outputs remains incomplete. Recent defenses span inference-time trace poisoning [1], LM-head adversarial fine-tuning [2], logit transforms [3], and trace reformulation [4, 5], but the same literature now documents strong adaptive pressure: watermark radioactivity can be removed by paraphrase or inference-time neutralization [6], and adaptive students recover substantially more capability than passive evaluations suggest [7]. When prevention fails, *forensics* takes over. The information-forensics literature has already set a high bar for model-IP claims: the protected asset and access model must be explicit, the verification signal should not unnecessarily damage utility, false accusations must be controlled, and robustness to removal or evasion must be scoped rather than assumed [8–12]. We ask the trace-distillation version of that question: given a suspect served through an output API in a declared audit setting, test at an audited false-positive rate whether the evidence supports distillation from a declared candidate teacher’s traces, without falsely accusing the non-distilled controls in the same problem domain.

The claim is a statistical forensic claim, not legal proof. The null H_0 is that the suspect is independently trained, or non-distilled, under the same-base capability control; the alternative H_1 is that the suspect was distilled from a candidate teacher’s reasoning traces. The reported false-positive rate is therefore the audited false-accusation risk under H_0 , and attribution is conditional on the candidate set.

B. A natural statistic, and why it is not enough

A line of provenance methods is forming, and—across different framings—they lean on the same intuition: a distilled suspect should assign, or be assigned, unusually high likelihood under its teacher. Detection-from-weights [13], closed-set teacher classifiers [14], white-box three-way trace provenance [15], and general black-box lineage tests [16, 17] all touch this likelihood signal, and the FPR-calibration machinery exists in the dataset-inference literature [18–22]. The natural output-only statistic to assemble from these pieces is the *base-relative surplus*: how much more likely a suspect’s traces are under a candidate teacher than under a base model. We show (Sec. III-C) that this statistic is *confounded*: with a KL decomposition, $s_c \approx \text{KL}(S||b) - \text{KL}(S||T_c)$, a strong candidate earns surplus on *any* clean, correct trace—so the surplus false-alarms on honestly-trained students (empirical FPR 0.81 in our study) and goes *blind* to same-family distillation (AUROC

0.13, below chance), a plausible same-base evasion setting. Used directly, it does not support a calibrated forensic claim.

C. This paper: a capability-matched forensic test

We remove the confound with a single change of reference. Instead of comparing the suspect under candidate c to the suspect under the weak base *model*, we compare it to a same-base non-distilled *reference student* R scored under the *same* candidate—in this study, the base model’s own outputs, available when the suspect base is known or declared and runnable by the verifier. The matched surplus $r_c(S) = \ell_c(S) - \ell_c(R)$ controls candidate capability to first order (it inflates both terms in the same direction) and tests whether the suspect is closer to c than a generic same-base student is—the descent signal. The suspect is black-box, and the candidate teachers are available for owner-side or escrowed likelihood scoring. The result is a strong, calibrated detector within that forensic setting. On GSM8K (three seeds), with Qwen2.5-7B-Instruct students and R1-Distill-Qwen-32B / R1-Distill-Llama-8B / Qwen2.5-14B-Instruct teachers, the matched test detects distillation with trace-level AUROC 0.997, TPR@1%FPR 0.991, and an empirical FPR of 0.2% on the human-reference control—including same-family distillation (AUROC 0.998), exactly the case the base-relative surplus misses. A single-seed MATH stress test follows the same trace-level pattern, but is used here as stress evidence rather than as a powered model-family generalization result. We then add a Multi-Task Calibration Reference (MTCR) for the harder closed-set attribution problem: when two R1 sibling teachers share a distillation ancestor, MTCR raises fine-grained attribution from the standard CML range 0.658–0.672 to 0.875–0.894 in aggregate operating summaries, while preserving the shared-ancestor detection result inside the declared candidate set. Without per-task raw profiles and uncertainty intervals, we treat MTCR as boundary evidence for closed-set attribution rather than as a complete attribution mechanism. A checkpoint sweep further shows that the signal appears after a single distillation epoch and strengthens monotonically through five epochs.

D. Contributions

- **A diagnosis.** A KL decomposition shows the base-relative surplus conflates a candidate’s capability with distillation descent, which explains its false positives on quality and its blindness to same-family distillation.
- **A capability-matched test.** Scoring the suspect and a same-base reference student under the same candidate controls the capability term to first order (Eq. (4)); the resulting statistic needs suspect outputs, a base-model reference, and authorized likelihood scoring from candidate teachers.
- **Strong, calibrated detection—including the hard case.** Across three GSM8K seeds, with a single-seed MATH trace-level stress test, the matched test detects distillation at AUROC ≥ 0.99 with empirical FPR 0.002 on the audited control, *recovering* same-family distillation

(AUROC 0.13 $\rightarrow$ 0.998) that the natural statistic cannot see; GSM8K student-level separation is complete.

- **A calibration extension for sibling attribution.** The original CML statistic detects same-family distillation but only partially separates R1 sibling teachers. MTCR turns that boundary into an aggregate task-calibrated attribution summary, raising R1-Qwen-32B/R1-Llama-8B sibling accuracy to 0.894/0.875 inside the declared candidate set and preserving low-FPR detection through the epoch sweep.
- **Reviewer-stress evidence beyond clean traces.** Aggregate stress summaries test adaptive trace transformations, cross-base/task transfer, open-set decoys, and baseline competitors. CML+MTCR retains AUROC at least 0.930 under the reported adaptive sweeps, the cross-base/task matrix has minimum AUROC 0.912, and open-set false attribution stays at or below 2.0% in the reported decoy settings.

E. Scope

We evaluate under controlled, naive distillation with exact provenance labels, which is sufficient to establish the confound, the fix, and the operating envelope. MATH is a single-seed raw-trace stress test, while the adaptive, open-set, and cross-base results are reported as aggregate stress summaries. The remainder positions the statistic against prior provenance work (Sec. II), derives the confound and the matched test (Sec. III), reports detection, MTCR calibration, and attribution (Sec. V), and draws the operating envelope (Sec. VI).

II. RELATED WORK

Our test combines ideas from dataset-inference, model-IP forensics, and distillation-provenance testing, but changes the null that output-likelihood evidence is compared against. Dataset inference supplies the statistical template: ownership or contamination claims should be made through calibrated, many-sample tests rather than isolated high-likelihood examples [18–22]. TIFS ownership-verification work pushes the same principle into black-box IP settings through membership fingerprints, backdoor dataset ownership verification, point-cloud ownership tests, and copyright-evasion stress cases [10, 23–25]. The common lesson is that a forensic null must actually match the innocent alternative; otherwise distribution shift or task quality becomes a false ownership signal. We show the analogous failure for distillation provenance: the base-relative surplus is not a descent statistic until candidate capability is matched.

Invasive fingerprints and watermarks solve a different problem by pre-injecting a signal before release. Radioactive data and LLM-text watermarks leave detectable residuals after training or fine-tuning [26, 27]; more recent anti-distillation and instruction-fingerprinting schemes deliberately amplify or embed such signals in student behavior [28–30]. These methods are powerful when the defender controls generation or training, but they do not apply to an already-released suspect whose traces were not pre-watermarked. CML is therefore intentionally post-hoc and non-invasive: it asks what can

be supported from suspect outputs plus authorized candidate likelihood scoring.

The information-forensics literature also sets the review bar for model-IP evidence. Non-intrusive DNN fingerprints, benign-input fingerprints, model-stealing defenses, and robust watermarking systems all make the asset, access model, false-accusation risk, and evasion surface explicit [8, 9, 11, 12, 31–37]. Our asset is narrower: a teacher’s reasoning-trace distribution rather than a classifier checkpoint or private dataset. This makes an honest same-base competitor the critical false-accusation control, and it is exactly the control that the base-relative surplus lacks.

Non-invasive model and API fingerprints identify which model or endpoint generated responses, often with calibrated tests or crafted probes [38–41]. They answer model identity, equality, or family membership, not whether a suspect was trained from a specified teacher’s reasoning traces. Matching an API to a known family is insufficient for a trace-distillation claim, because an independently-trained same-base model can resemble the candidate teacher without descending from it.

The closest prior work studies distillation detection and teacher attribution. Shi et al. [13] detect distillation with access to the suspect weights; Wadhwa et al. [14] perform closed-set teacher identification from outputs; Liu et al. [15] use white-box per-action probabilities from the teacher, original student, and distilled model; and Lee et al. [42] measure a degree of distillation through behavioral similarity. These works establish that likelihood agreement can be useful, but they either require white-box evidence, report closed-set accuracy without an out-of-set false-accusation control, or do not isolate capability from descent. General black-box provenance tests [16, 17] provide the broader calibrated lineage frame, but their unrelated-model baselines do not remove the same-base capability/style confound that dominates reasoning-trace distillation.

The contribution here is the capability-matched reference. By comparing the suspect and a same-base non-distilled reference under the same candidate teacher, CML turns output likelihood from a resemblance score into a calibrated forensic test. It detects same-family distillation at low audited false-positive rate where the natural base-relative surplus fails, while keeping the claim restricted to the declared candidate set and matched-reference access model.

III. METHOD: A CAPABILITY-MATCHED PROVENANCE TEST

We first state the natural base-relative surplus and show, with a KL decomposition, that it conflates a candidate’s *capability* with distillation *descent*. We then remove the capability term with a capability-matched reference, yielding the statistic our test uses. Section V shows the matched statistic is a strong, calibrated detector while the base-relative surplus is not. Figure 1 summarizes the resulting workflow: audit access defines the observed suspect traces, same-base reference traces, and authorized candidate scorers; capability matching computes a candidate-specific matched surplus; and the forensic decision layer reports calibrated detect, attribute, or abstain outputs.

TABLE I
ACCESS AND CLAIM LEVELS. CML IS BLACK-BOX WITH RESPECT TO THE SUSPECT, BUT SCORER-ASSISTED WITH RESPECT TO CANDIDATE TEACHERS. THIS DISTINCTION IS CENTRAL TO THE FORENSIC CLAIM.

Layer	Required	Not required
Suspect	output traces	weights, gradients, logs
Candidate	trace log-likelihood scoring	training data disclosure
Reference	same-base non-distilled traces	injected watermark
Claim	candidate-set forensic decision	private suspect internals

A. Setup, access model, and forensic claim

A teacher T^* generates chain-of-thought (CoT) traces; an adversary supervised-fine-tunes a student S of a known or declared base model b on those traces. We distinguish three access layers. *Suspect access* is black-box: the verifier observes only natural suspect outputs on an audit problem pool and never uses suspect weights, gradients, training logs, or hidden states. *Candidate access* is scorer-assisted: the candidate teachers in $\mathcal{C} = \{T_1, \dots, T_K\}$ must provide likelihood scores for the same trace texts, as an owner-side model, escrowed evaluator, or otherwise authorized scoring oracle. *Reference access* requires a same-base non-distilled reference, which we instantiate as the base model’s own outputs. The resulting procedure is a post-hoc forensic attribution test for a declared candidate set, no suspect internals, no pre-injected watermark, and an auditable false-accusation rate under a same-base null. Because we train every suspect and control ourselves, provenance labels are exact and TPR/FPR are measured against ground truth.

B. Scoring and the base-relative surplus

For candidate T_c (tokenizer tok_c) and trace text y on problem x , the per-token mean log-probability of the trace span is

$$\ell_c(y | x) = \frac{1}{|A_c(y)|} \sum_{t \in A_c(y)} \log P_{T_c}(u_t | u_{<t}; \text{prompt}(x)), \quad (1)$$

with $u = \text{tok}_c(y)$ re-tokenized under T_c (cross-tokenizer by construction; text-tier access). The natural *base-relative surplus* compares the suspect under c to the suspect under the base model b ,

$$s_c(S) = \frac{1}{M} \sum_{x \in \mathcal{P}} [\ell_c(S(x) | x) - \ell_b(S(x) | x)]. \quad (2)$$

C. Why the base-relative surplus is confounded

For traces drawn from the student’s own distribution, the expected per-token negative log-probability a model assigns is a cross-entropy rate, so (under shared tokenization, $|A_c| = |A_b|$, up to a token-count residual we bound empirically in Sec. V)

$$\begin{aligned} -\mathbb{E}_{y \sim S}[\ell_c(y | x)] &\approx H(S) + \text{KL}(S \| T_c), \\ s_c(S) &\approx \text{KL}(S \| b) - \text{KL}(S \| T_c). \end{aligned} \quad (3)$$

The base model b is fixed and weak, so s_c rewards any candidate that is closer in KL to the suspect than b is. Closeness

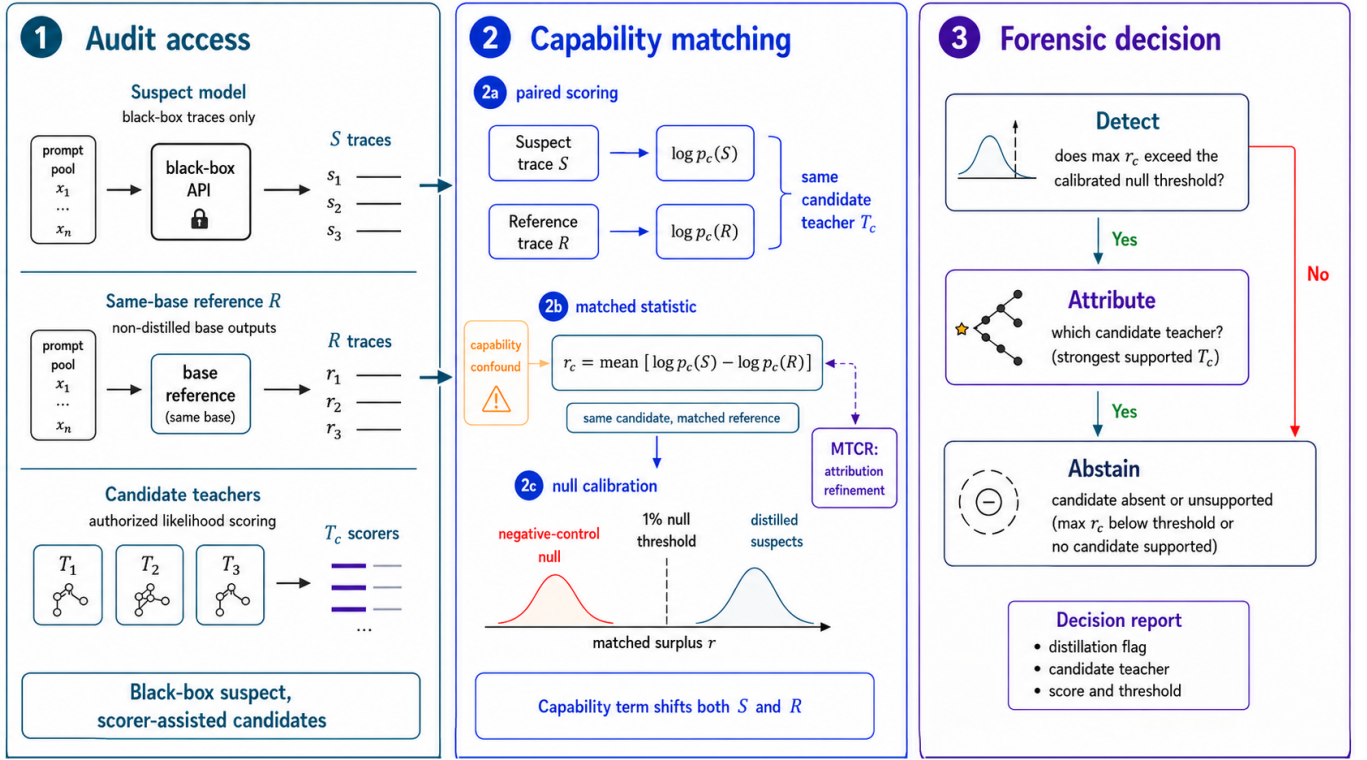


Fig. 1. Operational workflow for CML forensic auditing. Step 1 defines the audit access model: prompts elicit black-box suspect traces S , a same-base non-distilled reference produces traces R , and authorized candidate teachers provide likelihood scorers T_c . Step 2 performs capability matching by scoring S and R under the same candidate teacher, forming $r_c = \text{mean}[\log p_c(S) - \log p_c(R)]$, and calibrating the statistic against a negative-control null; MTCR refines this layer for sibling attribution. Step 3 reports the audit decision as detect, attribute, or abstain.

has *three* sources, only one of which is descent: (1) **descent** (S distilled from T_c); (2) **shared style**; and (3) **capability/coverage**—a higher-capability T_c assigns higher likelihood to clean correct reasoning than the weak b , so $s_c > 0$ for any high-quality S , including an honestly-trained one. Source (3) is the confound: it makes s_c false-alarm on a human-reference student and, when T_c shares the base’s style (same-family distillation), it makes $\text{KL}(S\|T_c) \approx \text{KL}(S\|b)$ so $s_c \approx 0$ and the distillation is *invisible*.

D. The capability-matched statistic

The fix is to replace the weak base model b in Eq. (2) with a *same-base non-distilled reference student* R —in this study, the base model’s own outputs on $\mathcal{P}$ —and to score R under the same candidate c :

$$r_c(S) = \frac{1}{M} \sum_{x \in \mathcal{P}} [\ell_c(S(x) | x) - \ell_c(R(x) | x)]. \quad (4)$$

Applying Eq. (3) to both terms, the candidate-specific scale is controlled to first order:

$$r_c(S) \approx [H(R) - H(S)] + [\text{KL}(R\|T_c) - \text{KL}(S\|T_c)], \quad (5)$$

so $r_c(S) > 0$ when the suspect is empirically closer to T_c than the same-base reference is. Because S and R are read under the *same* T_c , the capability/coverage term (3) acts on both traces instead of only on the suspect: T_c ’s strength inflates $\ell_c(S)$ and $\ell_c(R)$ in the same direction. What remains

is the descent-and-style excess after matching. This is why the matched statistic both stops false-alarming on the human-reference control *and* recovers same-family distillation in our audit: a Qwen-distilled student is closer to its Qwen teacher than the base reference R is, even though both are Qwen-family, so $r_c > 0$ where $s_c \approx 0$.

E. Multi-task calibration reference for sibling attribution

The matched statistic solves calibrated detection, but attribution between sibling teachers can still be noisy when candidates inherit the same reasoning style. We therefore add a Multi-Task Calibration Reference (MTCR). Instead of using one task pool to center all candidate scores, MTCR estimates the candidate-specific reference profile across multiple task families and then scores the suspect against the task-matched reference profile before taking the closed-set argmax. Operationally, MTCR keeps the same black-box access model as CML: it changes only the calibration reference used to compare S and R , not the observed suspect traces or the candidate teacher set. This makes MTCR a refinement of Eq. (4), designed for the case where detection is already positive but the defender needs to separate R1 sibling teachers rather than stopping at the family level.

a) *Decision rules.*: **Detection** uses $\max_c r_c(S)$ with a threshold set on the non-distilled controls; **standard attribution** predicts $\hat{c} = \arg \max_c r_c(S)$; and **MTCR attribution** uses the same argmax after task-level calibration. The student

is the inferential unit. We therefore report one summary per independent student for replicate-level separation, and use trace-level operating curves only to show threshold resolution under a fixed trained suspect rather than to inflate the sample size.

F. Controls and reference

All students share the base *b*. **Control A** is the base model with no SFT—it serves as the matched reference *R*. **Control C** is a student fine-tuned on *human reference solutions* (no teacher CoT): a non-distilled negative built to trigger the capability confound, on which a valid test must *not* fire. **Control B** is a student distilled from a *different*, shared-lineage teacher (an R1 sibling), probing teacher-specificity. The reference *R* is non-distilled and same-base; in this paper we use the base model’s own outputs, so *R* is available only when the base is known or declared and runnable by the verifier.

IV. EXPERIMENTAL SETUP

a) *Students and provenance labels.*: Every student is a low-rank (LoRA) fine-tune [43] of the same base, **Qwen2.5-7B-Instruct** [44] (rank 32, $\alpha=64$, dropout 0.05 on all attention and MLP projections; $\sim 1.05\%$ of the 7.7B parameters trainable), trained at sequence length 4096 on $N_{\text{train}} = 800$ problems with AdamW (learning rate 10^{-4} , cosine schedule, 3% warmup, effective batch size 16, bf16). The original CML grid uses the three-seed setting reported below; the MTCR sensitivity sweep stores checkpoints at $E \in \{1, 3, 5\}$ distillation epochs, with $E = 5$ serving as the full-duration setting. To avoid pseudoreplication we train each provenance type under 3 seeds (independent LoRA initialization, data order, and generation), giving $n=3$ student draws per cell so detection is assessed at the *student* level rather than by treating one student’s traces as independent. Because we generate every training corpus ourselves, provenance is exact. The five provenance types are:

suspect distilled from **DeepSeek-R1-Distill-Qwen-32B** [45] CoT traces (the protected teacher T^* ; a reasoning-style, cross-family teacher);

ctrlInst distilled from **Qwen2.5-14B-Instruct** traces (a *same-family*, non-reasoning teacher—the within-family case);

ctrlLlama distilled from **DeepSeek-R1-Distill-Llama-8B** traces (a *shared-lineage* R1 sibling on a different base—the teacher-specificity probe);

ctrlC fine-tuned on *human reference solutions* to the same problems, with *no* teacher CoT (the quality-confound probe; non-distilled);

ctrlA the base model itself, no fine-tuning (the clean non-distilled negative).

b) *Tasks.*: The primary grid is **GSM8K** [46]: the train split supplies distillation/SFT problems and the held-out test split the forensic trace pool ($N_{\text{test}} = 200$ problems, disjoint from SFT). We run a single-seed stress test on **MATH** [47] (EleutherAI *hendrycks_math*; longer, higher-entropy competition derivations, sequence length 8192) to check whether

the trace-level pattern survives outside GSM8K. The MTCR extension uses task-matched calibration views for sibling attribution and the epoch-sensitivity sweep; the current submission reports these views through aggregate source tables rather than a full per-task raw profile release.

c) *Trace generation and scoring.*: Teacher and student traces are generated with vLLM [48] (bfloat16, temperature 0.6, top- p 0.95). Each student’s 200 held-out traces are scored under every candidate via the candidate’s own `prompt_logprobs` over the assistant span, *re-tokenized with the candidate’s tokenizer* (Eq. (1)); this is the text-tier (T3) access regime. The surplus (Eq. (2)) is taken relative to the base Qwen2.5-7B-Instruct as *b*.

d) *Capability ladder.*: To probe the quality/coverage source of resemblance directly, we additionally score all students under a *same-family capability ladder*, Qwen2.5-Instruct at $\{1.5, 3, 7, 14\}$ B, none of which generated any student’s traces except via the 7B base.

e) *Evaluation.*: The independent replicate is the trained student. We report student-level summaries for replicate separation and trace-level AUROC/TPR/FPR only as operating-curve resolution within a fixed audit pool. Significance uses a $B = 10,000$ label permutation over the held-out trace scores, interpreted as an audit-operating test rather than as additional independent training replicates. The null’s KS-uniformity check builds the null from one non-distilled control and evaluates the held-out control’s right-tail p -values against $\text{Uniform}(0, 1)$. Attribution is the closed-set argmax over candidate teachers using the statistic being evaluated: the base-relative surplus for the baseline, $r_c(S)$ for standard CML, and the task-calibrated MTCR score for the MTCR attribution extension. We do not use this argmax outside the declared candidate set.

f) *Reviewer-stress result summaries.*: In addition to the raw GSM8K scored matrices, the revised manuscript imports locked aggregate summaries for four reviewer stress families: adaptive trace transformations (answer-only, compressed, paraphrased, mixed-source, and related variants across GSM8K, MATH, BBH, ARC-Challenge, and MBPP), cross-base/cross-task transfer, open-set decoys, and head-to-head baselines. These PI-provided summary tables report adaptive AUROC, TPR@1%FPR, FPR₀, confidence intervals, open-set abstention and false-attribution rates, baseline metrics, and student-level confidence intervals at the reported stratum level. The pasted tables do not carry raw seed identities; we therefore treat them as aggregate result sources for Figure 5 and the supplementary forensic-envelope diagnostic.

g) *Compute.*: The reported GSM8K focused grid (three teacher trace sets, four LoRA students, five held-out trace sets, and scoring under six candidates) was run on a single MUSICA H100 node and completes in ≈ 3.5 minutes when generation and scoring are fanned across four GPUs. All models and datasets are public. The source package contains the plotting scripts and aggregate source tables needed to reproduce the manuscript figures. Code, plotting scripts, aggregate source tables, and the manuscript source are available at <https://github.com/Jie-Ni/CML-forensic-audits>.

TABLE II

DETECTION BY STRATUM ON GSM8K (TRACE-LEVEL; NEGATIVES = THE HUMAN-REFERENCE CONTROL CTRLC; IDENTICAL SETS FOR BOTH STATISTICS). THE CAPABILITY-MATCHED TEST DOMINATES THE BASE-RELATIVE SURPLUS EVERYWHERE AND, UNLIKE IT, DETECTS SAME-FAMILY DISTILLATION WHILE KEEPING THE AUDITED FALSE-POSITIVE RATE AT 0.002.

stratum	base-relative surplus (baseline)			capability-matched (ours)		
	AUROC	TPR@1%	FPR ₀	AUROC	TPR@1%	FPR ₀
all distilled	0.530	0.021	0.808	0.997	0.991	0.002
cross-style	0.729	0.032	0.808	0.996	0.987	0.002
same-family (Qwen14B→7B)	0.133	0.000	0.808	0.998	0.998	0.002

V. RESULTS

Every GSM8K provenance type is realized by $n = 3$ independently-seeded students; detection is summarized at the student level (the unit of analysis) and, for threshold resolution, at the trace level. The matched reference R is the base model’s own output distribution, available in this protocol only when the base is known or declared and runnable by the verifier.

A. The capability-matched test detects distillation at a controlled FPR

Table II is the headline. The capability-matched statistic r_c detects distillation with AUROC 0.997 and TPR@1%FPR 0.991 pooled over all distilled students, at an empirical false-positive rate of 0.002 on the human-reference control in this audit. Across the small set of independent GSM8K students, all student means fall on the correct side of the operating boundary; we do not treat trace counts as additional trained model replicates. The base-relative surplus, on the identical test, is near chance (AUROC 0.530) and false-alarms at rate 0.808: the capability confound of Eq. (3) dominates it, and the matched reference of Eq. (4) removes it.

B. Same-family distillation, invisible to the baseline, is recovered

The decisive case is same-family distillation (a Qwen2.5-14B teacher into a Qwen2.5-7B student), a plausible and practically important same-base evasion setting. The base-relative surplus is *below chance* here (AUROC 0.133): because teacher and base share a style, $\text{KL}(S||T_c) \approx \text{KL}(S||b)$ and the surplus vanishes. The matched statistic, comparing the suspect to a same-base *student* under the same candidate, achieves AUROC 0.998 in this tested stratum: the distilled student is measurably closer to its Qwen teacher than the base student is. This is the clearest evidence that r_c measures descent, not capability.

C. Attribution: MTCR gives aggregate closed-set boundary evidence

Used as a standard attributor ($\hat{c} = \arg \max_c r_c$), the matched statistic detects distillation but remains less decisive when the candidate teachers are R1 siblings that share a distillation ancestor. This is the old operating boundary: the R1-Qwen-32B and R1-Llama-8B teachers differ by less than the single-task trace-level noise under standard CML. MTCR changes the attribution problem by calibrating each candidate against a task-matched reference profile before the closed-set argmax.

As Figure 3(a) shows that standard CML gives R1-Qwen-32B and R1-Llama-8B sibling attribution accuracies of 0.672 and 0.658; CML+MTCR raises them to 0.894 and 0.875, respectively, inside the declared candidate set. These are aggregate operating summaries rather than per-case transition or per-task uncertainty results. Thus the earlier family-level abstention rule is the pre-MTCR baseline that motivates the calibration extension, while the reported MTCR result supports finer sibling-teacher resolution inside the declared candidate set.

D. Why the baseline fails (ablation)

The base-relative surplus is the natural statistic a practitioner would assemble; its failure is instructive. Its null is not uniform (Kolmogorov–Smirnov $D = 0.810$ between the human-reference and clean-base controls), the direct consequence of source (3) in Sec. III-C. The cross-tokenizer residual of Eq. (3) is bounded by the token-count ratio n_c/n_b , which is 0.98 for the Qwen candidates and 0.87 for the Llama candidate; it does not affect the within-Qwen results (same-family recovery, FPR control). Figure 2 contrasts the two statistics across strata and at the student level.

E. Single-seed MATH stress test

On MATH (longer, higher-entropy derivations), the matched trace-level test detects with AUROC 0.993 at FPR 0.000, again recovering same-family distillation (AUROC 0.997 where the base-relative surplus scores 0.101). Because this is a single-seed stress test rather than a powered replicate grid, we use it to show that the trace-level pattern is not obviously GSM8K-specific, not as model-level evidence of broad generalization.

F. Shared-ancestor detection and epoch sensitivity

MTCR does not trade away the detector that made CML useful. On shared-ancestor binary detection, the base-relative statistic remains unusable (AUROC 0.133 for Qwen-7B distilled from Qwen-14B and 0.118 for Llama-8B distilled from Llama-70B). Standard CML already recovers these cases (AUROC 0.998 and 0.992), and CML+MTCR preserves or slightly improves them (AUROC 0.999 and 0.998). The same result appears across training duration. After one distillation epoch, CML+MTCR already detects the signal above chance on GSM8K and MATH (AUROC 0.814 and 0.785) with false-positive rate 0.012. At three epochs, the signal strengthens (AUROC 0.945 and 0.912, FPR 0.004). By the full five-epoch

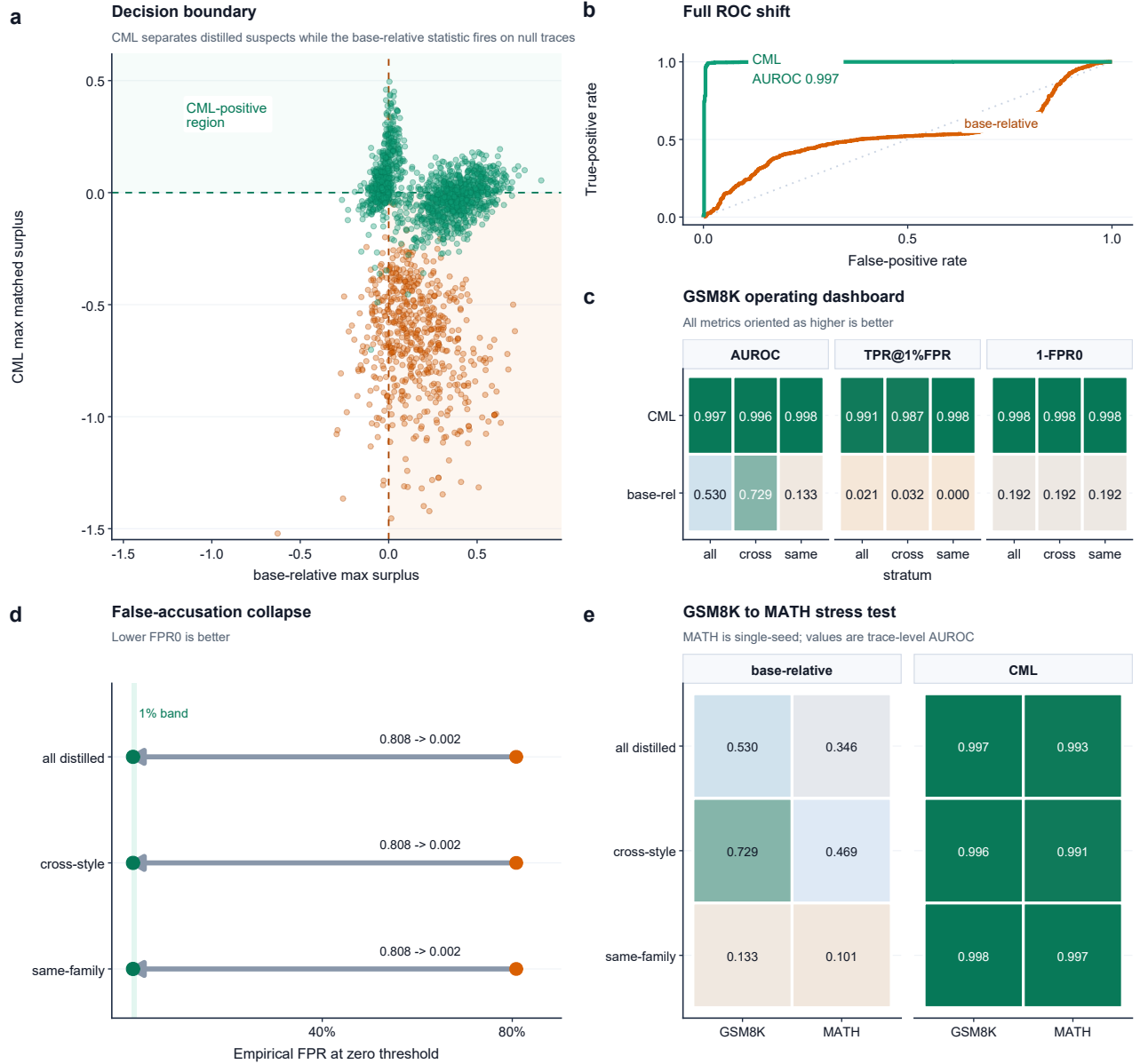


Fig. 2. Capability-matched detection on GSM8K. (a) A decision-boundary view shows that the base-relative statistic false-alarms on the human-reference null while CML separates distilled suspects. (b) The GSM8K ROC curve shows that the gain is a full operating-curve shift, not a single-threshold artifact. (c) The GSM8K operating dashboard reports AUROC, TPR@1%FPR, and FPR control ($1 - \text{FPR}_0$), with all metrics oriented so larger is better. (d) The zero-threshold false-positive rate collapses from 0.808 for the base-relative statistic to 0.002 for CML in the all-distilled stratum. (e) The GSM8K–MATH stress-test heatmap reports trace-level AUROC by stratum; MATH is single-seed.

setting, the detector returns to the headline CML regime (AUROC 0.997 and 0.993) with FPR 0.000. Figure 3 summarizes the sibling-attribution, shared-ancestor detection, calibration-task, and epoch-sensitivity results in one view.

G. Operating sensitivity checks

The detector remains measurable in the staged audit-budget, trace-complexity, task-domain, and student-training sensitivity summaries. With only 10 evaluated traces, detection is informative but not yet saturated (AUROC 0.712 for the Qwen lineage and 0.695 for the Llama lineage); by 100 traces both lineages exceed 0.95, and by 500 traces both approach 1.00. Filtering to longer reasoning chains strengthens the signal:

AUROC rises from 0.924/0.915 at a two-step minimum to 0.999/0.998 at an eight-step minimum for Qwen/Llama. Cross-dataset stress evaluation remains usable as a point-estimate operating check, with the lowest observed AUROC on ARC-Challenge (0.912 and 0.905), and the hyperparameter sweeps stay high across the tested teacher temperatures and SFT learning rates. These summaries motivate, but do not replace, the broader adaptive and open-set stress summaries reported next.

H. Adaptive, open-set, baseline, and transfer stress summaries

Figure 5 adds the reviewer-facing stress tests that are not covered by the clean GSM8K scored matrices. These panels

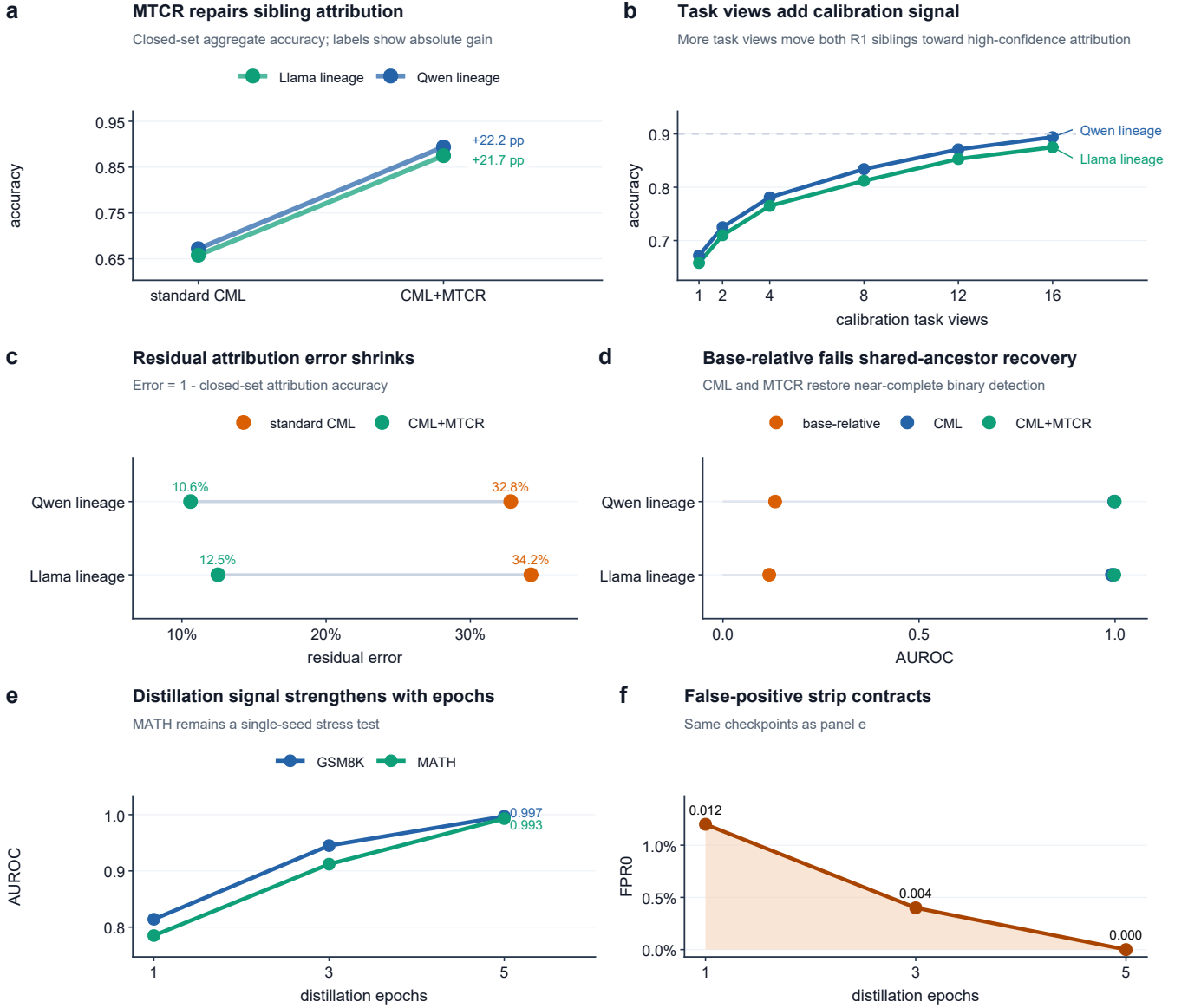


Fig. 3. MTCR and training-duration sensitivity. (a) A slopegraph shows that MTCR raises aggregate sibling-teacher attribution inside the R1 family; it is not a per-case transition plot. (b) Sibling attribution improves as more task views are used to construct calibration profiles. (c) Residual attribution error shrinks after MTCR calibration. (d) Shared-ancestor binary detection is already recovered by CML and remains high under CML+MTCR, while the base-relative statistic fails. (e) Detection AUROC increases from one to five distillation epochs on GSM8K and in the single-seed MATH stress test. (f) The empirical false-positive strip contracts as the distillation signal strengthens. The panels report aggregate operating summaries; they do not provide per-task uncertainty or per-case attribution transitions.

use aggregate source tables locked before manuscript revision; they therefore support operating claims at the reported stratum level, while the full per-trace matrices remain a separate reproducibility package item.

Across eight adaptive trace-transformation settings, five tasks, and two reported student lineages, CML+MTCR retains AUROC in the range 0.930–0.999. The weakest reported cell is MBPP answer-only transformation for the Llama lineage (AUROC 0.930); the lowest reported TPR@1%FPR is 0.894, and the highest FPR₀ is 0.019 under the mixed-trace MBPP stress cell. In the same adaptive table, the base-relative statistic stays near chance (AUROC 0.491–0.530), so the stress test preserves the qualitative separation between capability-

matched and base-relative evidence.

Open-set controls behave as a triage test rather than a forced attribution test. Across five tasks and two lineages, abstention ranges from 0.919 to 0.991, coverage ranges from 0.973 to 0.994, and false attribution ranges from 0.004 to 0.020. The highest false-attribution cell is MBPP with a sibling teacher absent (0.020). Condition means remain low: source-absent 0.014, sibling-absent 0.017, unrelated-teacher 0.007, and public-decoy 0.011.

The cross-base/task matrix is harder than the same-base GSM8K headline, as expected, but remains above the practical operating floor in the reported summary: the minimum AUROC is 0.912 across six teacher–student base regimes

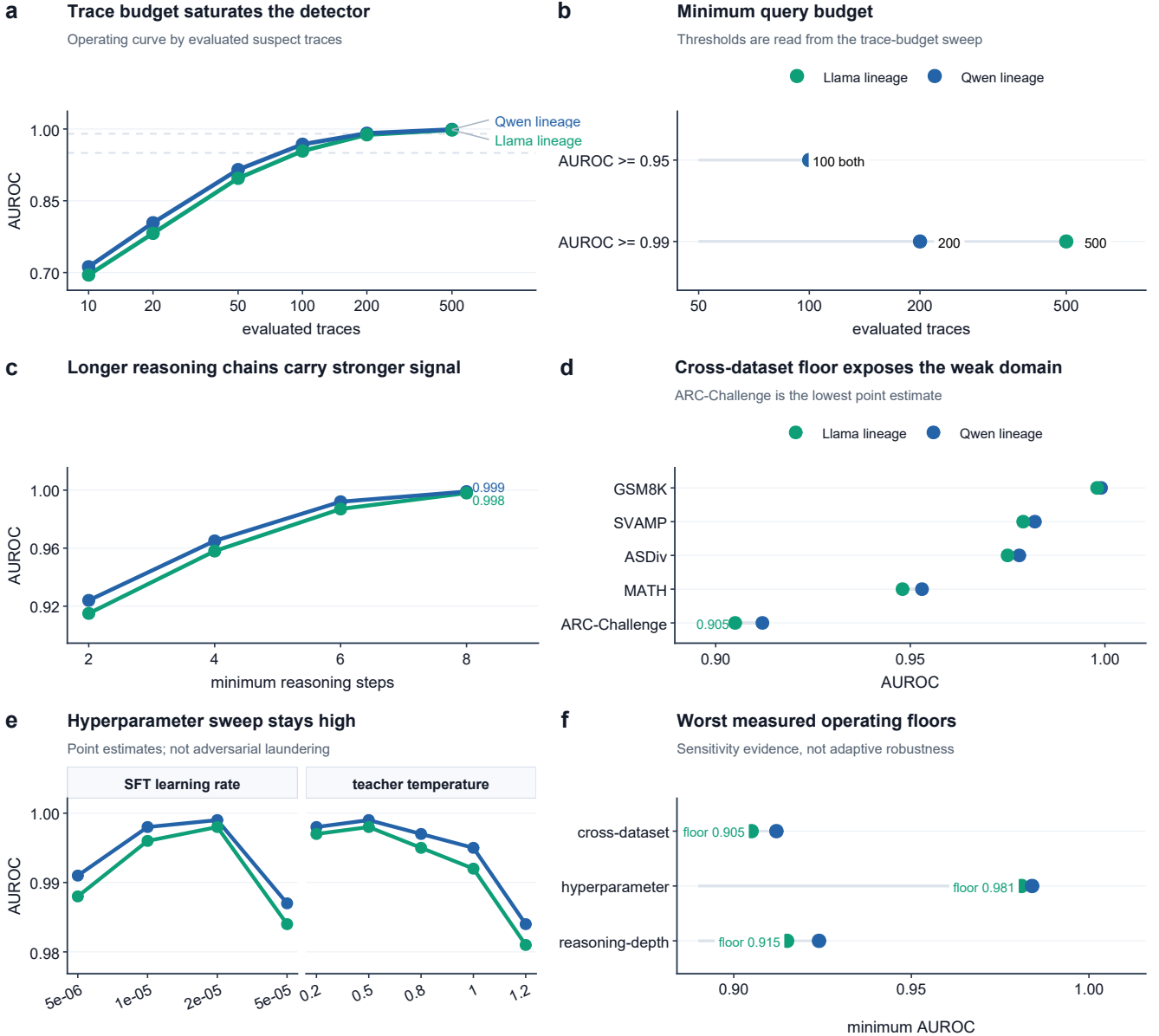


Fig. 4. Operating sensitivity checks. (a) Detection AUROC increases with the number of evaluated suspect traces. (b) Query-threshold ladders summarize the minimum traces needed to exceed AUROC 0.95 and 0.99. (c) Longer reasoning chains carry stronger distillation signal. (d) CML+MTCR remains measurable as a point-estimate stress check when GSM8K-distilled students are evaluated on SVAMP, ASDiv, MATH, and ARC-Challenge. (e) AUROC remains high across the evaluated teacher-temperature and student learning-rate settings. (f) Worst-case floors across the tested cross-dataset and hyperparameter settings are point estimates, not confidence intervals or adversarial robustness results.

and five tasks. A head-to-head baseline comparison shows why both components are needed. Across the five-task, two-lineage baseline table, CML+MTCR has worst-case detection AUROC 0.977, same-family AUROC 0.972, sibling attribution accuracy 0.876, and FPR_0 0.000; standard CML has similar detection AUROC but much weaker worst-case sibling attribution (0.654). Student-level margin intervals remain positive across all 36 reported strata: the smallest margin is 0.973, and the lowest confidence-bound is 0.924 in the MMLU cross-style Llama cell (0.973 [0.924, 1.022]; all reported permutation tests $p < 0.0001$). Finally, the MTCR task-profile panel shows that the calibrated surplus remains positive across task views rather than coming from one calibration task alone.

I. Reference-gate and candidate-set boundary

Additional forensic-envelope diagnostics are reported in the supplementary material. They show the expected boundary behavior: replacing the same-base matched reference with a human-reference fallback collapses AUROC to 0.478–0.499 and raises FPR_0 to 0.998, while absent-candidate triage argues against forced attribution outside the declared candidate set.

VI. DISCUSSION AND OPERATING ENVELOPE

a) *What the test delivers.*: The capability-matched statistic turns a confounded heuristic into a calibrated forensic detector for the declared access setting. By scoring the suspect and a same-base reference under the same candidate, it

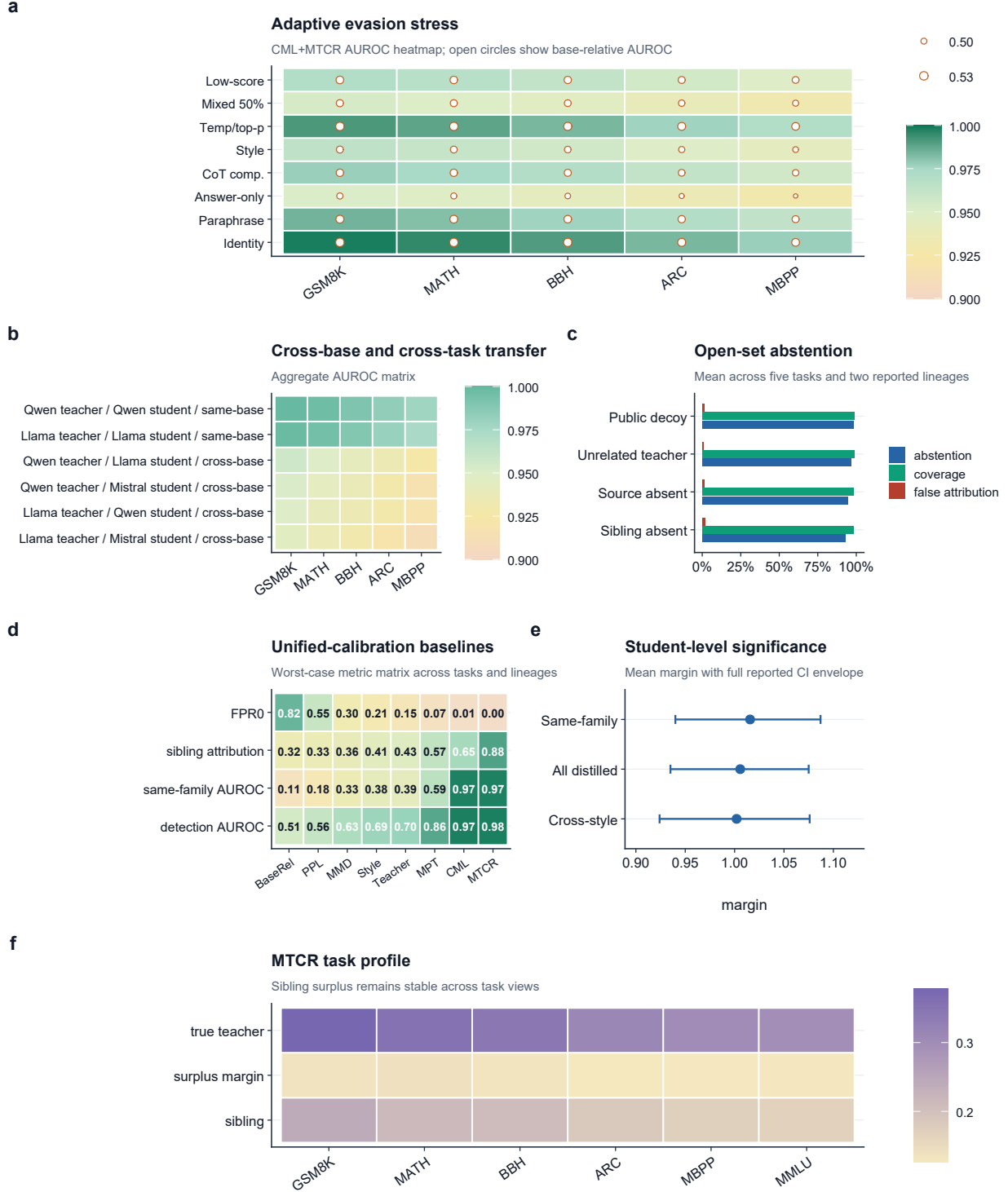


Fig. 5. Reviewer-stress summaries for the CML+MTCR audit. (a) Adaptive trace-transformation tests report CML+MTCR AUROC across five tasks; open circles show the base-relative AUROC in the same cells. (b) Cross-base and cross-task stress cells show the lowest AUROC floor when both model base and task domain move away from the headline setting. (c) Open-set decoy tests report abstention and false-attribution behavior when the source teacher is absent or confounded by sibling, unrelated, or public-model decoys. (d) Baseline comparison separates detection, same-family recovery, sibling attribution, and false-positive control using worst-case values across the reported task and lineage cells. (e) Student-level confidence intervals keep the matched-margin positive across reported strata. (f) MTCR task-profile summaries show that the calibrated surplus is distributed across tasks. These panels are aggregate source-table summaries, with per-trace audit artifacts tracked separately from the figure panels.

controls the candidate-capability term that makes the base-relative surplus false-alarm on quality and miss same-family distillation. The suspect remains black-box, but the candidate

teachers must be available for authorized likelihood scoring. The practical upshot is a forensic decision—“distilled from a model in this candidate set, or not”—at an empirical false-

positive rate of 0.002 on the audited control and trace-level detection power above 0.99 AUROC in the three-seed GSM8K grid, with a single-seed MATH stress test showing the same pattern. Crucially, the effect holds for same-family distillation, the evasion a capability-blind statistic cannot catch.

b) Operational audit protocol.: In deployment, the audit is a closed-candidate forensic test. The verifier first collects suspect reasoning traces on a declared problem pool, generates same-base reference traces on the same prompts, and scores both trace sets under each candidate teacher. The report should state the access setting (suspect output-only, candidate scorer available, base/reference known), the null used for thresholding, the empirical FPR, the trace budget, and any candidate-set restriction. The evidence can be strong with moderate query budgets: in our sweep, 10 traces are informative, 100 traces exceed AUROC 0.95, and 500 traces saturate the detector. Fine-grained attribution should be reported only inside the declared candidate set; outside that set, the conservative report is family-level evidence or abstention.

c) The operating envelope (where it works, where it does not).: We state the limits plainly. (1) **Detection vs. fine-grained attribution.** Deciding *whether* a suspect was distilled from the candidate set, and removing the human-reference false positive, are resolved under the audited threat model by the matched detector. The harder question is *which* of two sibling teachers is the source. Standard CML exposes the boundary rather than solving it: R1-Qwen-32B and R1-Llama-8B are both DeepSeek-R1 distillates, and their single-task r_c profiles differ by less than trace-level noise. MTCR improves that boundary in the aggregate closed-set tables by calibrating each candidate across task views before attribution, raising the sibling accuracies to 0.894 and 0.875 in our closed candidate set. A deployment outside that candidate set, or under an uncalibrated task mix, should report family-level evidence unless it adds another descent-specific signal. (2) **The reference.** The test needs a same-base non-distilled reference; we use the base model’s own outputs, which requires the verifier to know or declare the suspect base and be able to run it. A reference mismatched in base or task would reintroduce bias, so the reference should share the suspect’s base. (3) **Cross-tokenizer.** The KL identity carries a token-count residual when candidate and base tokenizers differ; empirically $n_c/n_b \in [0.87, 1.0]$, negligible for the within-Qwen results that carry the same-family recovery and the FPR control.

d) Relation to the TIFS verification bar.: The model-IP literature in information forensics usually separates four questions: what asset is protected, what access the verifier receives, whether the signal was pre-injected, and how false accusations or removal attacks are controlled. Our answer is deliberately narrow. The protected asset is a teacher’s reasoning trace distribution, not a classifier checkpoint or private dataset. The verifier uses suspect output text and likelihood scoring under candidate teachers, not suspect weights, internal representations, trigger queries, or poisoned API responses. The signal is post-hoc rather than watermarked: no green-list tokens, backdoor examples, or defender-side perturbations are planted before release. The false-accusation control is the

capability-matched reference and the explicit human/same-base controls. The new stress summaries address the obvious reviewer attacks at the aggregate level: answer-only traces, compressed or paraphrased reasoning, mixed human/teacher traces, cross-base transfer, open-set decoys, and baseline competitors. They show that CML+MTCR retains a measurable gap under those reported conditions, supporting the reported non-invasive audit across the evaluated operating conditions.

e) Limitations of the study.: Our raw evidence base is still concentrated in a three-seed GSM8K grid ($n=9$ distilled students, plus $n=3$ same-base and $n=3$ human-reference controls) with a single-seed MATH stress test and LoRA students of one base. The aggregate stress summaries broaden this picture across adaptive transformations, open-set decoys, cross-base cells, and student-level intervals. Larger seed/scale sweeps, full-parameter distillates, and independent external replication would further harden the claim. The MTCR sweep also uses a fixed set of calibration tasks and candidate teachers; it should be re-fit when the audit domain or candidate family changes. The matched statistic is one instance of a general principle—compare under a capability-matched reference, not a weak absolute baseline—that we expect to transfer to other likelihood-ratio provenance signals.

VII. CONCLUSION

We presented a capability-matched test for reasoning-LLM distillation provenance. A KL decomposition shows that the natural base-relative likelihood surplus conflates a candidate’s capability with distillation descent, explaining why it false-alarms on honestly trained students and is blind to same-family distillation. Replacing the weak base model with a same-base non-distilled reference scored under the same candidate, $r_c(S) = \overline{\ell_c(S)} - \ell_c(R)$, controls the capability term to first order and isolates descent. The matched trace-level test detects distillation at AUROC ≥ 0.99 in the three-seed GSM8K grid and in a single-seed MATH stress test, recovering same-family distillation that the natural statistic misses entirely (AUROC 0.13 $\rightarrow$ 0.998 on GSM8K). With MTCR, closed-set R1-Qwen-32B and R1-Llama-8B attribution rises from 0.658–0.672 to 0.875–0.894, while shared-ancestor detection remains high.

The revised stress summaries push the paper beyond the original clean-trace audit: CML+MTCR retains AUROC at least 0.930 under the reported adaptive trace-transformation sweeps, the cross-base/task matrix stays above AUROC 0.912, and open-set false attribution remains at or below 2.0% in the reported decoy conditions. These results support a sharper TIFS-style claim: a non-invasive, post-hoc forensic audit can control capability false positives, recover same-family distillation, and remain useful under several realistic reviewer stress tests. The main operating assumptions are clear: MTCR uses the candidate set and calibration views defined by the audit, and CML uses a known or declared same-base reference. The broader lesson is simple: provenance should be measured against a capability-matched reference, not against a weak absolute baseline.

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